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Experimental and numerical studies on the vortex-induced vibration of two-box edge girder for cable-stayed bridges

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A R T I C L E I N F O

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A B S T R A C T

Due to the blunt aerodynamic configuration, the edge-girder bridge deck may sustain severe vortex-induced vibrations (VIVs) that adversely affect bridge usability and security. In the present study, the VIV performance and the related aerodynamic mechanism of the two-box edge girder for a cable-stayed bridge are systematically investigated. Based on the flow pattern and vortex structure around the section simulated by computational fluid dynamics, it is inferred that the aerodynamic interactions of the large-scale vortices under the deck and the shedding vortices in the wake zone periodically change the fluctuating pressures on the section and induce the vibrations of the deck. Moreover, several aerodynamic mitigation countermeasures are designed. Their effects on the vertical VIV are experimentally investigated in detail. It is found that the vertical VIVs are considerably suppressed at different attack angles ($-3°$, $0°$, and $+3°$) by installing the mini-triangular wind fairings, which can satisfy the bridge wind-resistant design requirements. Further numerical simulations show that after the installation of the effective countermeasures, vortex shedding in the wake zone is mitigated, and the strengths and distortions of large-scale vortices around the upper and lower surfaces of the section also decrease, thereby the vertical VIV is suppressed effectively.

1. Introduction

The steel and concrete composite bridge deck is considered to be an extremely competent type of medium-span bridge because of its good material properties and economic cost (Su et al., 2015). The edge girder section, which is among the most common form of composite bridge decks, is widely applied in cable-stayed bridge design. Based on aerodynamic standards, however, the edge girder bridge deck has an inadequate aerodynamic stability particularly against vortex-induced vibration (VIV) because of its bluff body. Although the VIV of bridge decks is not as dangerous as flutter, large amplitudes and continuous vibrations would affect the comfortability of the bridge and may even cause fatigue failure and structural damage. In the past, severe VIVs have occurred on several built edge girder bridges. The Kessock Bridge is a two-I-shaped steel edge girder bridge (main span: 240 m), which sustained a large-amplitude VIV when the wind speed was approximately 22 m/s (Owen et al., 1996). The Deer Isle-Sedgwick Suspension Bridge is a 2-H-shaped edge girder bridge, where vertical VIVs with four different modes occurred when the wind speed was in the range 9.11–15.9 m/s (Kumarasena et al., 1991). The Rio Niteroi bridge is a slender continuous steel edge girder bridge with twin-box girders that experiences frequent strong vertical oscillations caused by cross-winds with a sustained velocity of approximately 15.3–16.7 m/s. When a severe VIV occurred, it left people panic-stricken to the extent that some even abandoned their cars on the bridge (Battista et al., 2000). Evidently, the VIV should be carefully considered in the design of wind-resistant edge girder bridges.

Numerous researches related to the aerodynamic characteristics of VIV of bridge decks exist. The aerodynamic mechanism of VIVs, however, has not been rationally explained, and it remains considerably difficult to obtain theoretical solutions because of the complexity of fluid–structure interaction (Simiu and Scanlan, 1996; Ehsan et al., 1990; Zhu et al., 2013; Sun et al., 2014). In bridge engineering, optimizing the aerodynamic shape of the bridge section in the preliminary stage of bridge design is a typical approach for improving wind resistance. Kubo et al. (2001) employed flow visualization technology to study the aerodynamic stability of the "п" section, and the VIV responses were mitigated by adjusting the width of the two longitudinal boxes. Moreover, edge girder sections with different shapes (i.e., "I", box, and circular) have been experimentally investigated to improve the wind-resistance performance (Daito et al., 2002). It is found that adjusting the longitudinal box slope

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affects the aerostatic response (Lee et al., 2017). Additionally, the installation of an aerodynamic passive attachment on the bridge deck is commonly employed in the design of wind-resistant bridges. Sakai et al. (1993) conducted a series of experiments to investigate the VIV performance of the edge girder section (composed of two "I" longitudinal beams) by setting various aerodynamic suppression countermeasures. They also recommended the use of a "tip plate" to effectively suppress VIVs. For another two-I-shaped edge girder cable-stayed bridge, the VIV was remarkably suppressed by installing a combination of vertical stabilizers and air flow depressing boards (Yang et al., 2015). The installation of vertical stabilizers on the upper and lower surfaces of the bridge deck is also commonly used measure to mitigate the aerodynamic instability caused by flutter (Zhou et al., 2018). It is known that vortices, which separate from the leading edge of bluff bodies, are three-dimensional(Li et al., 2020). And the aerodynamic forces generated by the fluctuating pressures is not completely correlated along span (Wilkinson, 1981; Sun et al., 2019). Disrupting the vortex consistency would diminish the vortex force along the bridge, thereby reducing the VIV amplitude (El-Gammal et al., 2007). For example, sealing the traffic barrier at certain intervals was successfully applied to suppress the vertical VIV of a streamlined box girder; however, it should be noted that sealing the traffic barrier increases the drag force on the deck (Li et al., 2018a). The CFD method has been widely used to investigate the aerodynamic mechanism of the rectangular configuration which is the basic cross-section in the researches of bridge decks (Pan et al., 2007; Wu et al., 2020). Moreover, to obtain the aerodynamic mechanism of VIV of a bridge deck, Xu et al. (2018) investigated the nonlinear dynamic stability of the VIV of an edge girder bridge deck by employing both CFD method and wind tunnel test. The CFD and particle image velocimetry methods are widely utilized in simulating the flow pattern around the deck to explore the VIV triggering mechanism and propose countermeasures for suppressing these vibrations (Sarwar et al., 2010; Vaz et al., 2016; Li et al., 2018b; Ma et al., 2018). Apart from aerodynamic countermeasures, mechanical countermeasures are also introduced to suppress VIVs (Chen et al., 2013). Among these is the installation of a tuned mass damper (TMD), which is one of the most effective techniques for controlling the VIV of long-span bridges (Gu et al., 2002).

Although numerous studies related to the VIV performance of the edge girder section have been performed, investigations on the aerodynamic mechanism of a two-box edge girder are limited and should therefore be explored. Furthermore, countermeasures with good economic cost and engineering applicability for this type of bridge deck should be proposed. In this study, to preliminarily understand the VIV mechanism of a two-box edge girder and propose effective mitigation countermeasures, a wide-width cable-stayed bridge is employed as an engineering example. First, a series of section model wind tunnel tests is performed to experimentally investigate the VIV at three different attack angles and damping ratios. Thereafter, the CFD method is employed to simulate the flow field around the deck to qualitatively analyze the VIV triggering mechanism of the edge girder deck. Finally, based on simulation results, systematic wind tunnel tests of mitigation countermeasures (i.e., sealing the rails and installations of apron plate, vertical stabilizer, horizontal flow-isolating plate (HFIP), and wind fairing) are performed to suppress the VIV.

2. Experimental setup details

2.1. Engineering background

The Yanpingba Yangtze River Bridge in China is a steel and concrete composite bridge with an overall length of 866 m and a span arrangement of 193 + 480+193 m. The main span deck, which is 40 m wide and 3.86 m high, is a typical edge girder section with two steel rectangular boxes. The two fish-shaped concrete towers are 182.9 m high. Based on the climatological data at the bridge site, the design mean wind speed at the height of the bridge deck is 34.9 m/s. The general arrangement of the bridge and standard section configuration are shown in Figs. 1 and 2, respectively.

2.2. Section model and experimental setup

The section model wind tunnel tests were performed in a smooth flow at the second test section of XNJD-1 wind tunnel of Southwest Jiaotong University. The tunnel's cross-sectional dimensions are 2.4 m (width) $\times$ 2.0 m (height) $\times$ 16.0 m (length). The wind speed can be continuously adjusted from 0.5 to 45.0 m/s, and the turbulence intensity is less than 0.5% of the corresponding wind speed.

The section model geometrical scale is chosen as 1:60 to satisfy the appropriate aspect ratio. The Design Manual for Roads and Bridges stipulates that the corrections are required if the blockage of the wind tunnel test section induced by the model and its immediate surroundings exceeds about 5–10% (HAE, 2001). In the present investigation, no blockage correction is applied because the maximum blockage ratio of the tests is 5.2%. The aerodynamic shape of the longitudinal box is simulated using high-quality wood to ensure that the model has sufficient stiffness. The sidewalk boards, handrails, traffic barriers, and inspection traveler rails are manufactured using plastic materials (ABS). The section model and the special bracket suspended by eight linear tension springs enable the system to vibrate vertically and torsional. Two thin plates are installed at both sides of the section model to ensure two-dimensional flow and render end effects. The equivalent mass and mass moment of inertia are simulated by counterweights. The VIV is extremely sensitive to the damping ratio; hence, viscous dampers are employed to adjust and simulate the vertical and torsional damping ratios of the system. Dynamic responses are measured by three laser sensors arranged outside the wind tunnel.

The main parameters, including geometric dimensions, mass characteristics, and fundamental frequencies, are listed in Table 1. The photographs of the section model are shown in Fig. 3, and the support system of the section model test is illustrated in Fig. 4.

3. VIV characteristic of original section

3.1. Wind tunnel test results of original section

Currently, there is no generally accepted method for estimating the damping ratio of a bridge before construction because the bridge structure has considerable complexity. For the steel and concrete composite bridge, 0.48% (logarithmic decrement is 0.03) and 1% are the suggested damping ratios in the Design Manual for Roads and Bridges (HAE, 2001) and in the Wind-Resistant Design Specification for Highway Bridges (CMT, 2018), respectively. Various damping levels of the dynamic test system are therefore selected to investigate the effects of damping on the VIV amplitude in the section model test. The dynamic parameters of the experimental system are identified by using the free vibration approach in quiescent air before the wind tunnel test is conducted. The VIV performance of the model at the lowest damping ratio (vertical, 0.32%; torsional, 0.26%) are first tested at three attack angles (-3°, 0°, and $+3^{\circ}$); test results are shown in Fig. 5. The amplitude and wind speeds are both converted to those of the prototype bridge.

The Reynolds number corresponding to experimental investigations is ranged from 2.1×10^3 to 3.4×10^4. The testing results indicate that significant vertical VIVs occur at three different attack angles. As shown in Fig. 5(a), two lock-in ranges appear at the $+3^{\circ}$ attack angle: one is in the range 8–12 m/s, and the other has a 400-mm peak amplitude in the range 18–35 m/s. As for the 0° attack angle, the first lock-in range is approximately 10–15 m/s, and second has a 250-mm peak amplitude in the range 17–25 m/s. Only one lock-in range, approximately 15–23 m/s, is found at the -3° attack angle, and the peak amplitude approaches 250 mm. Based on the comparison above, a good agreement between the VIV performances at 0° and -3° attack angles is observed. As shown in Fig. 5(b), the torsional VIVs also occur at three different attack angles; however, the onset wind speeds of VIVs are all greater than 25 m/s,

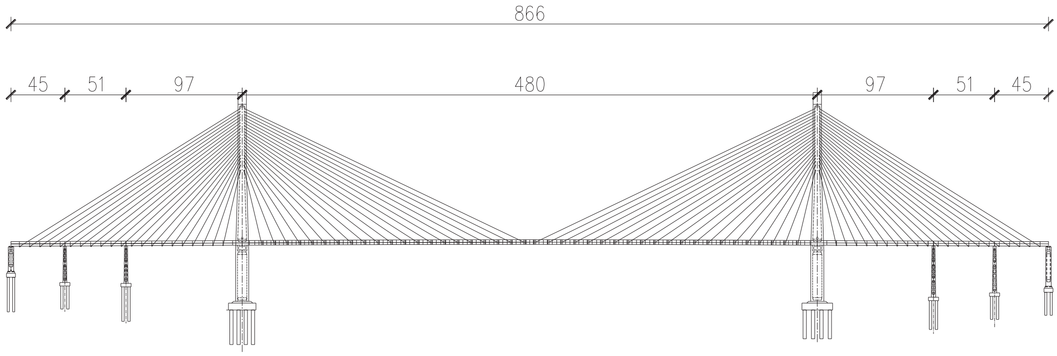


Fig. 1. General arrangement of bridge (unit: m).

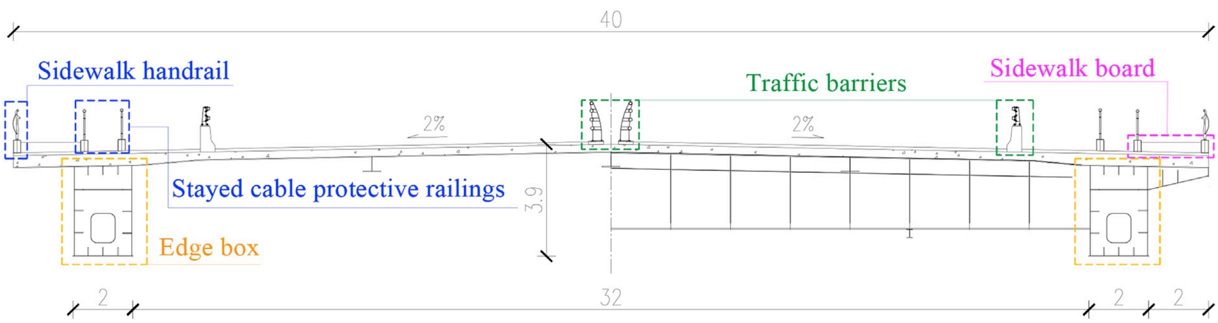


Fig. 2. Configuration of two-box edge girder section (unit: m).

which is rarely encountered at the bridge site, the comfortability of the bridge is not a problem because traffic is controlled. The vertical VIV should therefore be further investigated in detail, whereas the torsional VIV of the deck could be ignored. In order to study the vertical VIV performance at different damping ratios, further tests are conducted with damping ratios of 0.70% and 0.97%; test results are shown in Fig. 6.

Table 1
Parameters of wind tunnel tests.

Property	Unit	Scale	Prototype	Model
Length (L)	m	/	/	2.095
Height (H)	m	1/60	3.86	0.064
Width (B)	m	1/60	40	0.667
Mass per unit length (m)	kg/m	$1/60^2$	5.72×10^4	15.92
Mass moment of inertia per unit length (I_m)	$kg{\cdot}m^2/m$	$1/60^4$	7.28×10^6	0.564
Vertical frequency (f_h)	Hz	/	0.337	3.47
Vertical damping ratio (ξ_v)	%	/	/	0.32, 0.70, 0.97
Torsional frequency (f_α)	Hz	/	0.770	7.93
Torsional damping ratio (ξ_t)	%	/	/	0.26

According to the Chinese bridge wind-resistant design specification (CMT, 2018), the allowable vertical VIV amplitude can be calculated as follows:

$$h_a = \frac{0.04}{f_b} \tag{1}$$

where h_a is the vertical VIV allowable amplitude, and f_b is the frequency of the first vertical bending mode of the bridge. Based on Eq. (1), the allowable vertical VIV amplitude of the concerning bridge is 118.7 mm. Evidently, the vertical VIV responses of the deck at the three different attack angles cannot satisfy the allowable amplitude. Although tests at higher damping ratios are conducted, effective VIV mitigation countermeasures should be proposed to suppress the VIVs and ensure bridge usability.

Fig. 3. Section model in XNJD-1 wind tunnel.

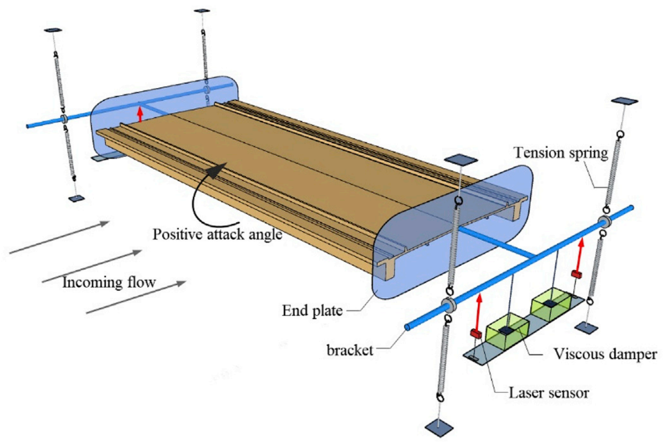


Fig. 4. Support system of section model test.

3.2. Numerical simulation of original section

Over the past few decades, the use of the CFD method in bridge engineering has rapidly developed. The CFD method can be used to simulate the flow pattern and vortex structure around the section. This can aid in qualitatively analyzing the VIV triggering mechanism, especially for determining the component and region of the section causing the VIV. Highly targeted aerodynamic suppression countermeasures that are economical and efficient can then be proposed based on subsequent wind tunnel tests and simulation results.

3.2.1. Numerical simulation setup

In this study, the two-dimensional flow field around the section is numerically simulated using commercial software ANSYS. The computational domain arrangement, which is composed of the section model zone and three rigid flow fields, is shown in Fig. 7. The multi-block meshing strategy is employed to generate a high-quality mesh with low computational complexity. The height of the first grid layer near the section is selected from derived values to guarantee that y+≤1 and to precisely simulate the flow separations. The mesh size increase ratio is approximately 1.05 from the inside out. The input velocity is consistent with that corresponding to the peak amplitude of the VIV wind tunnel test. The Reynolds numbers corresponding to the attack angles of -3°, 0°, and $+3^\circ$ are 2.6×10^4, 2.2×10^4, and 3.5×10^4, respectively. The numerical model of the flow around the bridge deck is formulated using the SST k-ω turbulence model.

3.2.2. Numerical simulation results of original section

The CFD simulation is first performed at the 0° attack angle. The Strouhal number depends on body geometry and Reynolds number. Under certain conditions, a stationary bluff body will shed alternating vortices according to the Strouhal relationship. The Strouhal number can be calculated by $S_t = f_s d/U$, where S_t is the Strouhal number, f_s is the frequency of full cycles of vortex shedding, d is the horizontal dimension of the bridge deck, and U is the mean wind speed of the incoming flow. Fig. 8 shows that the Strouhal numbers of the bridge deck in the numerical simulation at different attack angles are consistent with experimental values. It could be inferred that for this section, periodic vortex shedding is accurately simulated. As illustrated in Fig. 9, the lift coefficients of the section produced by numerical simulation exhibit a distinct periodicity.

The aerodynamic force acting on the model integrates the pressure contributions on the model surface depending on the distortion, shape and strength of vortices (recirculation zones), and their location on the model contour as well as on the distortion of surfaces exposed to the direct impingement by the main flow (Vaz et al., 2016). In order to visually depict the flow pattern and vortex structure, streamlines are adopted to describe the evolution of vortices (recirculation zones). The streamlines around the original section in one period at the 0° attack angle are shown in Fig. 10. The figure illustrates that the incoming flow significantly separates at the windward of the section, and the shear layers become unstable at the upper and lower regions of the deck. Along the downwind direction, a large number of vortices are generated at the upper and lower surfaces of section. A pair of vortices periodically sheds from the leeward edge in the wake zone. The upper vortex in the rear

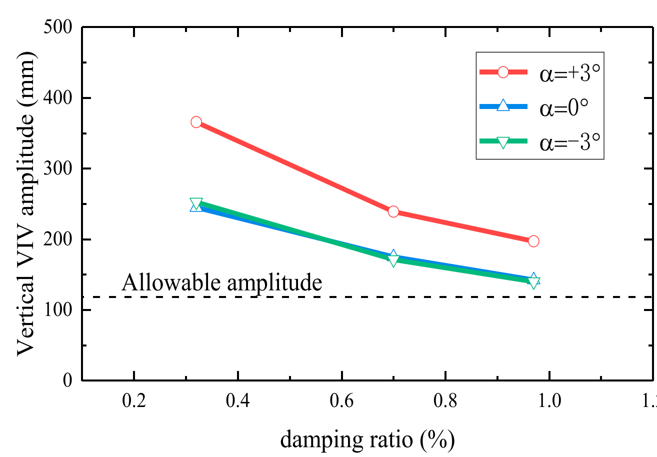


Fig. 6. Vertical VIV amplitudes at three different damping ratios.

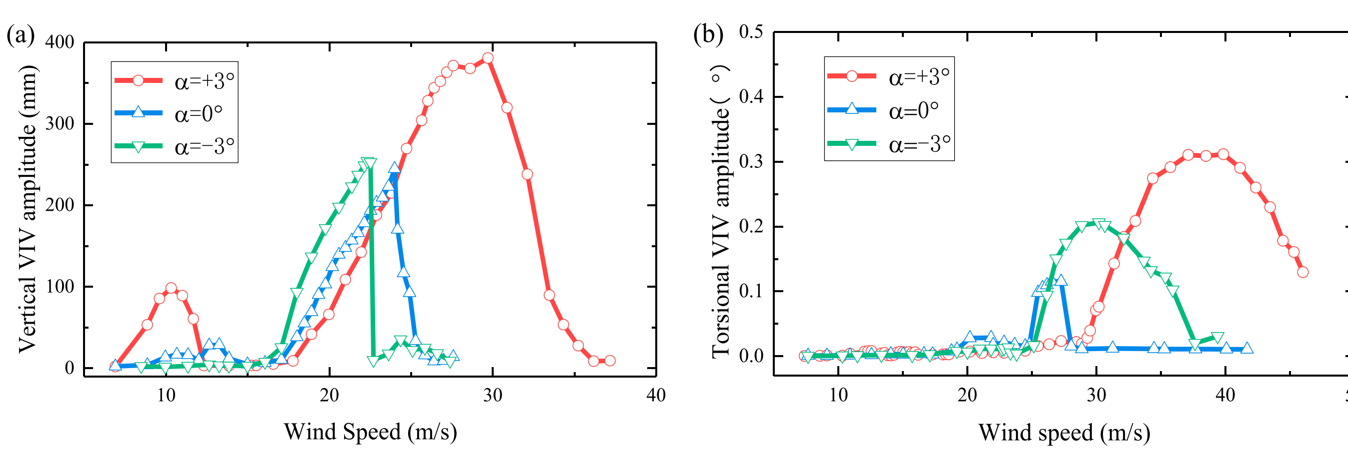


Fig. 5. Vertical and torsional VIV amplitudes of the original section.

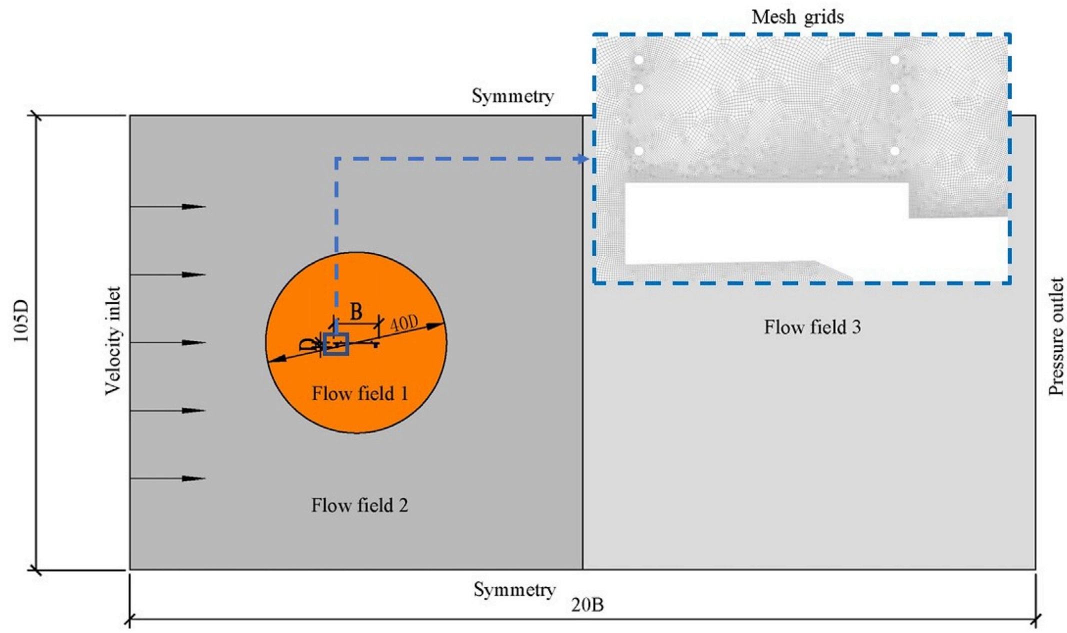


Fig. 7. Layout of computational domain.

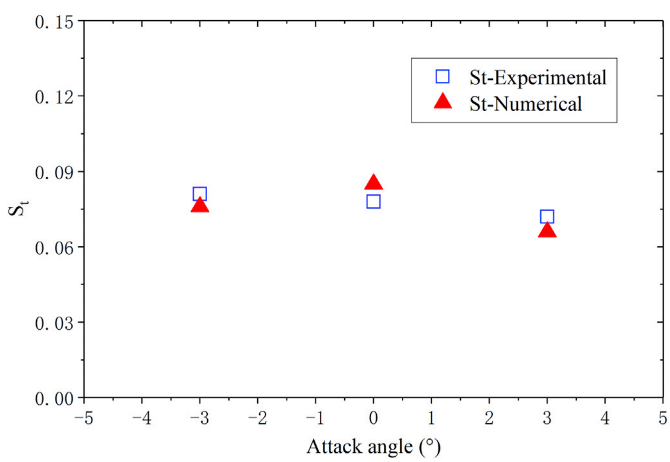


Fig. 8. Comparison of numerical and experimental Strouhal numbers.

region rotates clockwise and is generated at $t = 0$ and $t = T/8$. Over time, it moves along the downwind direction with the scale gradually increasing from $t = T/8$ to $t = 5T/8$ at the leeward edge of the deck. Finally, the upper vortex gradually weakens and eventually disappears. The lower vortex rotates counterclockwise and exists throughout the entire cycle. Meanwhile, the large-scale vortices exist beneath the deck. The streamline diagrams illustrate that the single core of the large-scale vortices would first split into two groups and then merge into one in the process of the vortex shedding in the rear region of the deck. The aerodynamic interactions of the shedding vortices in the wake zone and the large-scale vortices under the deck periodically change the pressure on the section surface and induces deck vibration. If the vortex shedding frequency approaches that of the bridge deck, the vortex shedding process would be captured by the deck oscillation; the deck movement might intensify the interaction and energy conversion. This aerodynamic instability is called lock-in phenomenon, and the periodic oscillation is the VIV. Vaz et al. (2016) and Li et al. (2019) studied other bridge sections with bluff body aerodynamic characteristics by applying the CFD method and found a similar VIV triggering mechanism.

4. VIV mitigation countermeasure research

4.1. Preliminary countermeasure design and tests

In the previous section, the simulation of the flow pattern and vortex structure around the deck by the CFD method and the preliminary analysis of the aerodynamic mechanism of the VIV are presented. The VIV may be caused by the following: aerodynamic instability induced by the incoming flow that significantly separates at the windward side of the edge girder section; the distortion of large-scale vortices at the upper and lower regions of the deck; the periodic shedding of vortices in the wake zone downstream of the section. It could be inferred that there are three regions that causes the VIV: the windward region, lower region, and rear region of the deck. Based on the aerodynamic characteristics obtained by numerical simulation, the countermeasures are divided into four categories according to the positions where vortices are generated and the vibration control mechanisms. These categories are railing optimization, installation of apron plate, installation of vertical stabilizer, and optimization of the aerodynamic shape of the two edge boxes. All these countermeasures are extensively tested by section wind tunnel tests. Thirteen representative cases are selected; the classification of aerodynamic attachments is summarized in Table 2. First, the VIVs of the developed sections at a damping ratio of 0.97% and an attack angle of 0° are investigated through wind tunnel tests; test results are shown in Fig. 11.

4.1.1. Railing optimization

Some researchers claim that the incoming flow would separate at the affiliated structures attached to the bridge (e.g., handrail, traffic barrier, and inspection traveler railing). It is presumed that this might produce fluctuating pressures that intensify the VIV of the bridge deck. As shown in Fig. 12, Li et al. (2018a) proposed an efficient method to suppress the VIV by reducing the spanwise correlations of vortex-induced forces through sealing the railing at intervals. For the bridge under consideration, based on the results shown in Fig. 11, sealing the outside handrails at intervals (Case 1) would not reduce the VIV, whereas sealing the protective railings (second railing) of the stayed-cable at intervals can slightly suppress the VIV. It should be noted, however, that sealing the railings would affect the aerostatic performance of the bridge deck because it may increase the drag force of the wind load.

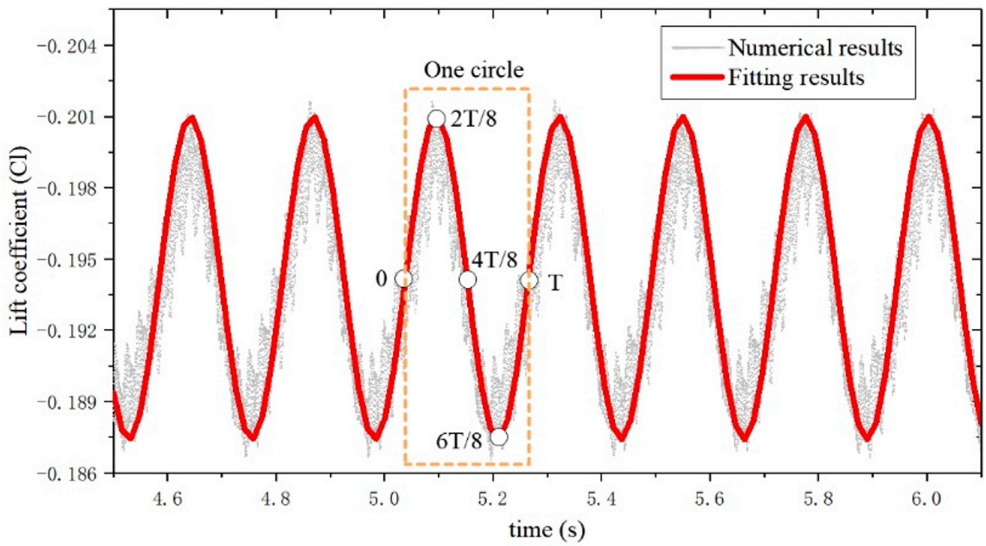


Fig. 9. Lift coefficients of numerical simulation.

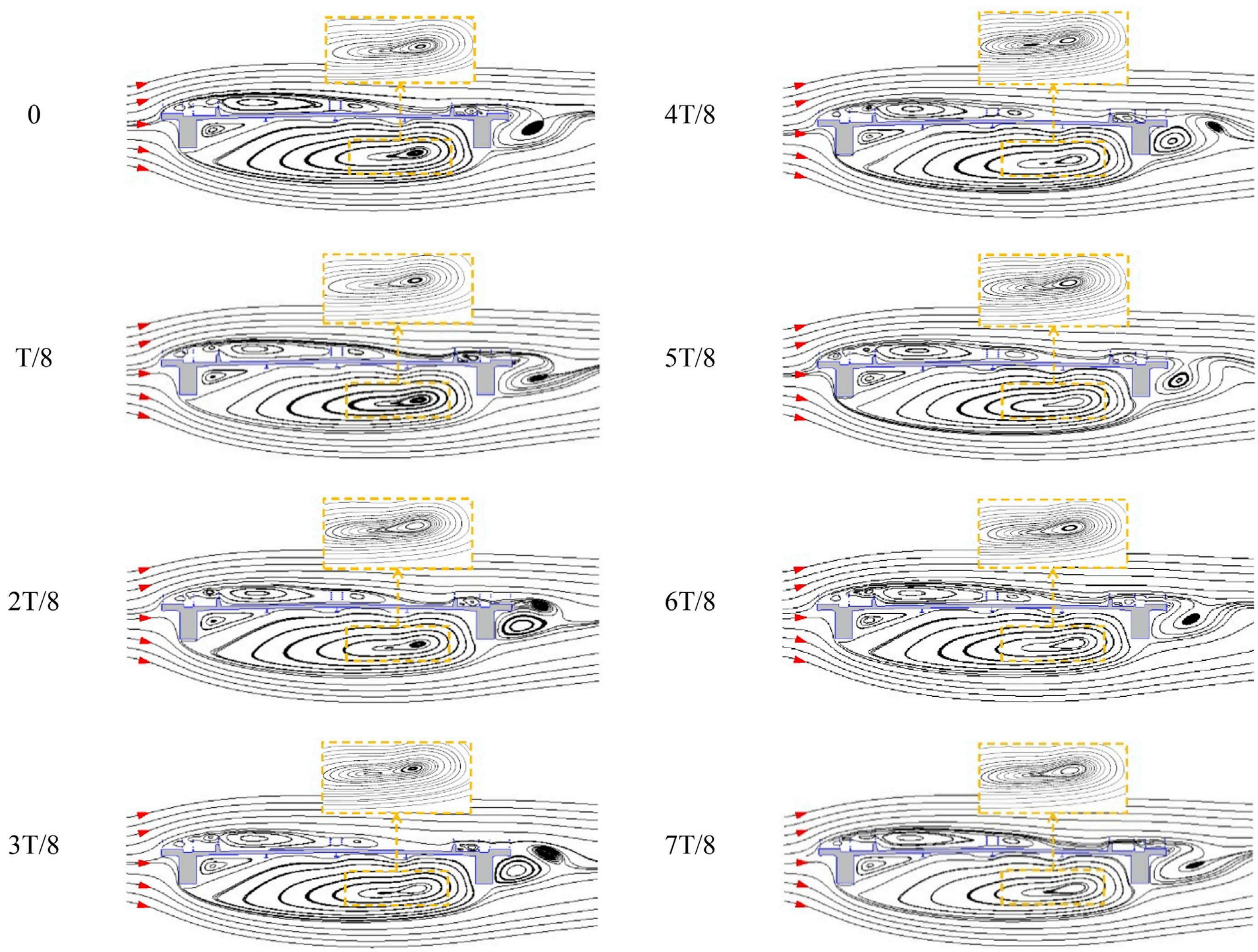


Fig. 10. Numerical streamlines around static section at 0° attack angle.

Table 2
Classification of 13 VIV countermeasures.

Categories of countermeasure design	Case number
Railing optimization	Cases 1 and 2
Installation of apron plates	Cases 3, 4, and 5
Installation of vertical stabilizer plates	Cases 6, 7, and 8
Optimization of aerodynamic shape of edge boxes	Cases 9, 10, 11, 12, and 13

4.1.2. Installation of apron plate

As depicted in Fig. 10, the incoming flow significantly separates in the windward region of the edge girder section and causes aerodynamic instability. According to the proposal of Sakai et al. (1993), the installation of apron plates on a two-I-shaped edge girder could mitigate the flow separation from the windward edge of the section and prevent the growth of separation bubbles that produce excitation forces. This type of countermeasure has been successfully employed in Shanghai Nanpu Bridge and Qingzhou Minjiang Bridge in China. For the two-box edge girder in this research, three different countermeasures ("tip plate" (Case 3), "L" edge plate (Case 4), and "I" edge plate (Case 5)) are designed based on the apron plate shape. The test results show that although the "tip plate" installation is an effective suppression measure on the two-I-shaped edge girder section, it would increase the VIV responses (Sakai et al., 1993). For the "L" edge plate in Case 4, a new lock-in range is observed at a lower wind speed; hence, although the VIV amplitude of the original section at a higher wind speed is reduced, it should be ignored. After the installation of the vertical "I" edge plate (0.52H high) in Case 5, the VIV of the deck is significantly suppressed, and the vibrations practically disappear.

4.1.3. Installation of vertical stabilizer

Vertical stabilizers are commonly used for improving the aerodynamic stability in the design of wind-resistant bridges. Many studies indicate that the installation of vertical stabilizers at the upper and bottom surfaces of the deck considerably alters the vortex form and flow pattern around the section, significantly improving the VIV and flutter performance (Sakai et al., 1993; Chen et al., 2006; Zhou et al., 2018). For the bridge under consideration, when a central vertical stabilizer is installed under the deck (Case 6), the VIV amplitude approximates that of the original section. In addition, the installation of a central vertical stabilizer on the upper surface would not mitigate the VIV (Case 7). If three parallel vertical stabilizers are installed under the deck (Case 8), then the VIV performance will be remarkably improved, and the VIV amplitude would satisfy the Chinese bridge design specification (CMT, 2018).

4.1.4. Optimization of aerodynamic shape of edge box

With respect to aerodynamic standards, the edge box is a typical bluff body, and the sharp corners at the windward and leeward sides will generate a large number of periodic vortices that may induce vibration. By optimizing the aerodynamic shape of the edge box (Cases 9–13), the incoming flow around the section could be smoothed, and flow separation and vortex shedding would be reduced, which may mitigate the VIV. In Case 9, the deflectors installed at regular intervals under the edge boxes could reduce the spanwise correlations among vortex-induced forces and improve the aerodynamic performance. Such a type of deflector has been applied to suppress the VIV in the case of Hanjiatuo Yangtze River Bridge (a trussed-girder bridge) in China. For the bridge under consideration, however, the VIV of the deck is slightly suppressed (Case 9 in Fig. 11). The installation of wind fairing is also among the most common aerodynamic countermeasures in wind-resistant design, especially for streamlined box girders. This countermeasure allows the incoming flow to pass the deck smoothly, thus improving the deck's aerodynamic performance. The inclined angles and dimensions of wind fairings are the key factors that mitigate vibrations (Haque et al., 2016; Li et al., 2018b). The two measures in Cases 10 and 11 (Fig. 11) are traditional integral wind fairings, based on the test results, however, the wind fairings in Cases 10 and 11 do not improve the VIV performance of this deck. This is probably because of the large aspect ratio ($B/H = 10.4$) and the inappropriate form of wind fairings. For Case 12, the test results show that if a mini-triangular wind fairing is installed on the edge box, the VIV can be significantly suppressed. Li et al. (2019) indicated the dominant pressure fluctuation at the lower leeward surface of the section would be reduced by installing horizontal flow-isolating plates (HFIP), which can suppress the vortex-induced force and improve the aerodynamic stability of an edge girder bridge deck. In Case 13, the HFIPs arranged close to the bottom plate of the edge boxes suppress the VIV amplitude by approximately 35%.

In summary, according to the preliminary test results, several countermeasures can effectively suppress VIVs: the sealing of the stayed cable protective railing (second railing) at intervals and the installation of vertical "I" edge plate, three vertical stabilizers, mini-triangular wind fairing, and HFIP. The dimensions and cost of the countermeasures in previous tests, however, are not optimal; hence, it is necessary to propose more reasonable countermeasures based on the above preliminary tests.

4.2. Countermeasure optimization experiment

By considering the engineering applicability and cost, the vertical "I" edge plate, mini triangle wind fairing, and HFIP are selected for further optimization and investigation. The optimization experiments are conducted at attack angles of -3°, 0°, and $+3^\circ$ and a damping ratio of 0.97%.

4.2.1. Vertical "I" edge plate

The effectiveness of vertical "I" edge plates with five different dimensions as countermeasures is tested; their heights are 0.20H, 0.25H, 0.30H, 0.35H, and 0.45H. The test results shown in Fig. 13 show that the VIV amplitudes gradually decrease with the increase in edge plate height at 0° and -3° attack angles. At these attack angles and when the height of the "I" edge plate reaches 0.35H, the VIV phenomenon practically disappears. This type of edge plate, however, is not effective if the attack angle is $+3^\circ$.

4.2.2. Mini-triangular wind fairing

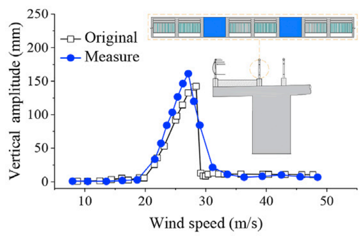
The test results in Case 12 show that the VIV is significantly suppressed by installing mini-triangular wind fairings with dimensions of $\beta = 45^\circ$and $h = 0.18H$ on the edge boxes. In order to obtain a more reasonable measurement, several triangular wind fairing dimensions are tested. As shown in Fig. 14(b), the triangular wind fairings have significant effects on the VIV at three different attack angles. For Cases 12–1, the vibration at the $+3^\circ$ attack angle is more severe than that in the original section, and the triangular wind fairing measurement in Case 12–3 most effectively suppresses the VIV. By installing such a wind fairing, the VIV amplitudes could well satisfy the Chinese bridge design specification at 0° and -3° attack angles; the VIV amplitude at the $+3^\circ$ attack angle is reduced by 50% and is just below the maximum allowable amplitude.

4.2.3. Horizontal flow-isolating plate (HFIP)

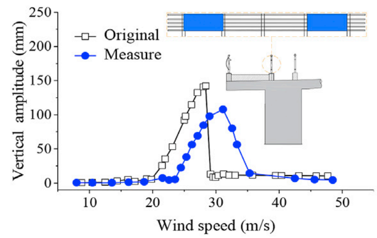
The HFIPs with five different widths, i.e., 0.20H, 0.28H, 0.36H, 0.44H, and 0.52H, are tested as countermeasures; test results are shown in Fig. 15. It can be observed that the five different HFIP sizes can significantly improve the VIV performance at the 0° attack angle, but the VIV amplitudes are only reduced by approximately 10% at the -3° and $+3^\circ$ attack angles. Moreover, their effectiveness as countermeasures depends on the HFIP width. It is found that 0.44H is the optimal HFIP dimension to suppress vibration.

4.3. Numerical simulation of improved section

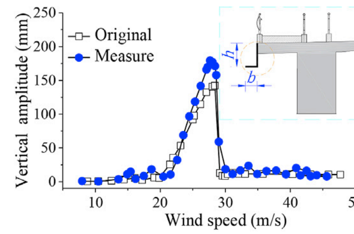
As shown in Figs. 13 and 14, the VIV of the two-box edge girder is efficiently suppressed by installing vertical "I" edge plates and mini-triangular wind fairings, indicating that the aerodynamic characteristic



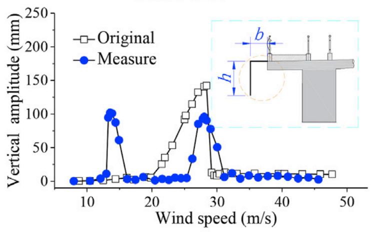
Case 1: Sealing outside railing at intervals



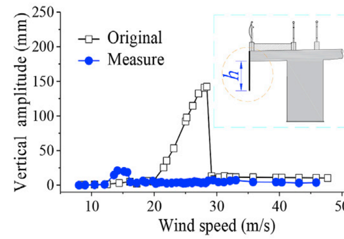
Case 2: Sealing second railing at intervals



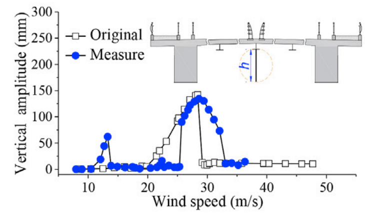
Case 3: Tip plate ($b = 0.2H$, $h = 0.3H$)



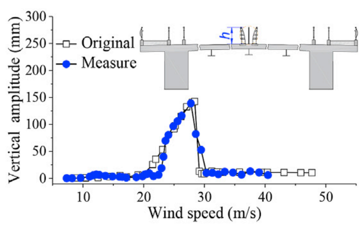
Case 4: "L" edge plate ($b = 0.15H$, $h = 0.52H$)



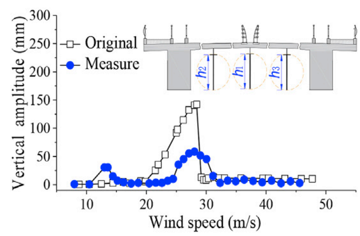
Case 5: "|" Edge plate ($h = 0.52H$)



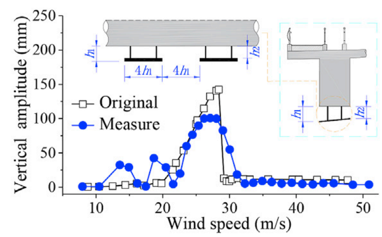
Case 6: Central vertical stabilizer ($h = 0.90H$)



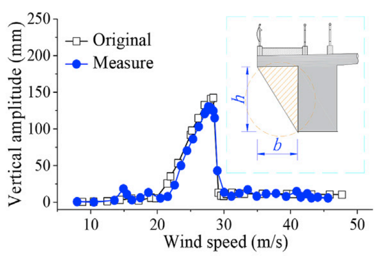
Case 7: Vertical stabilizer on the upper surface ($h = 0.35H$)



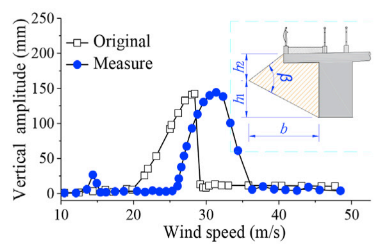
Case 8: Three vertical stabilizers ($h_1 = 0.90H$, $h_2 = h_3 = 0.82H$)



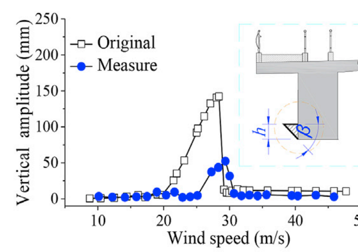
Case 9: Deflector ($b = 0.2H$, $h_1 = 0.26H$, $h_2 = 0.21H$)



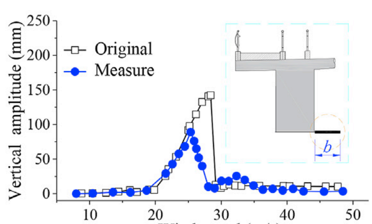
Case 10: Sealing cantilever ($b = 0.52H$, $h = 0.93H$)



Case 11: Traditional wind fairing ($\beta = 64°$, $b = 1.04H$, $h_1 = 0.40H$, $h_2 = 0.55H$)



Case 12: Mini-triangular wind fairing ($\beta = 45°$, $h = 0.18H$)



Case 13: Horizontal flow-isolating plate (HFIP) ($b = 0.40H$)

Fig. 11. Test results of VIV countermeasures (attack angle: 0°, damping ratio: 0.97%); *H* represents deck height.

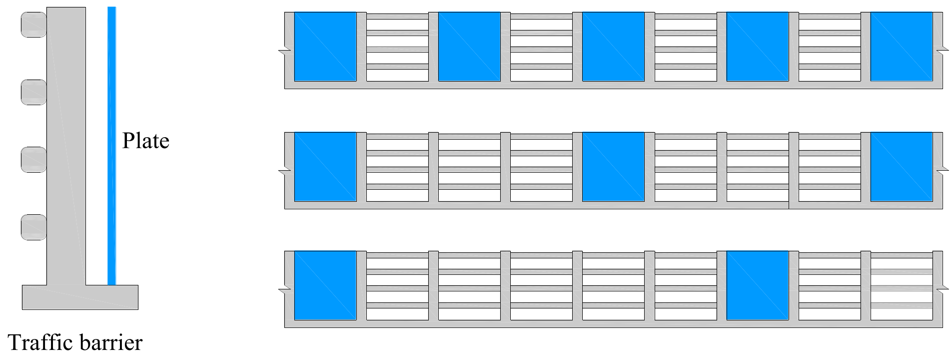


Fig. 12. The diagrams of sealing traffic barriers at intervals (Li et al., 2018a).

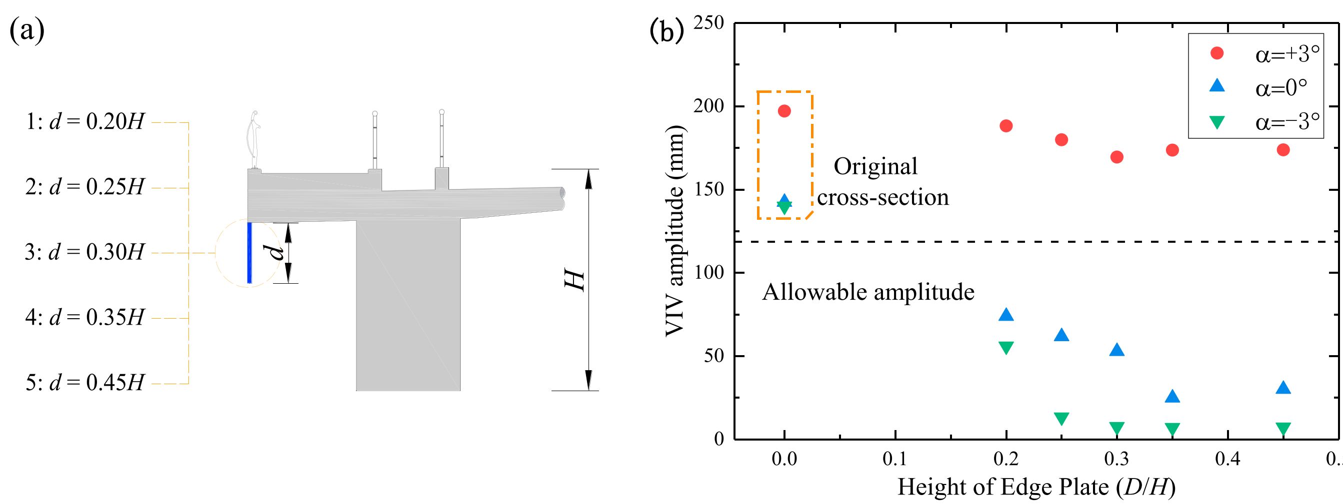


Fig. 13. Vertical VIV responses of deck with different vertical “l” edge plates.

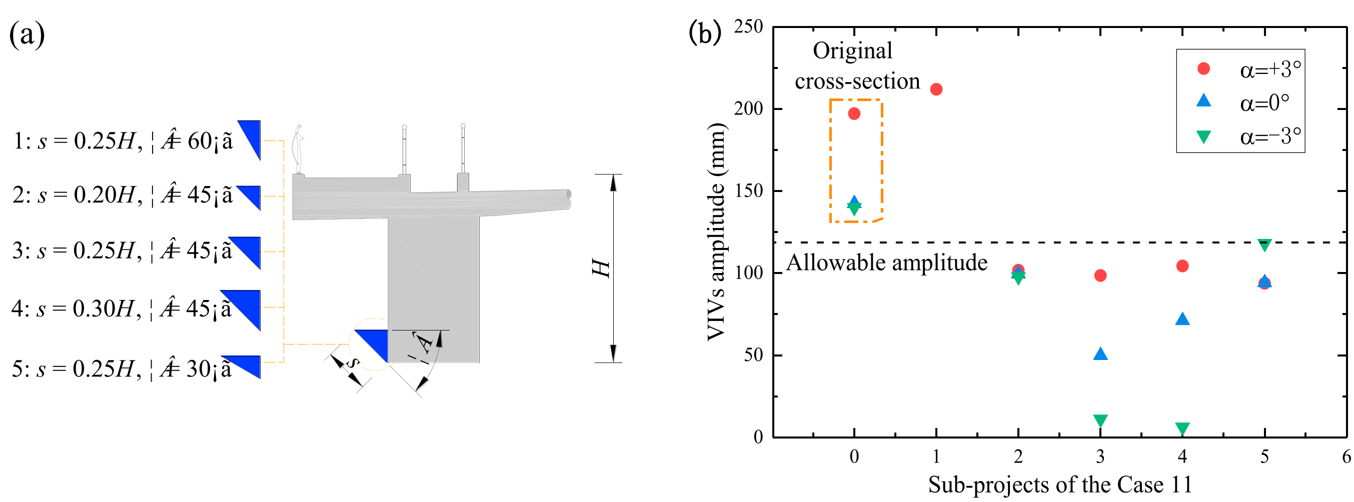


Fig. 14. Vertical VIV responses of deck with different triangular wind fairings.

of the section is remarkably changed by such measures. Compared to the triangular wind fairing, however, the vertical “l” edge plate is not effective at the +3° attack angle. It can be inferred that when these two countermeasures are installed, the effects of the different attack angles on the flow pattern and vortex structure remarkably differs. In order to comprehensively investigate the aerodynamic mechanism of countermeasures, the flow patterns around the section at attack angles of 0° and +3° are selected for further numerical simulations; results are shown in Figs. 16 and 17, respectively.

Based on the analysis in Section 3.2, the interactions of shedding vortices in the wake zone and the large-scale vortices under the deck probably induces the periodic vibration of the bridge deck. Fig. 16 clearly illustrates that there is a stable vortex at the wake region throughout the entire cycle after the installation of the triangular wind fairing, whereas

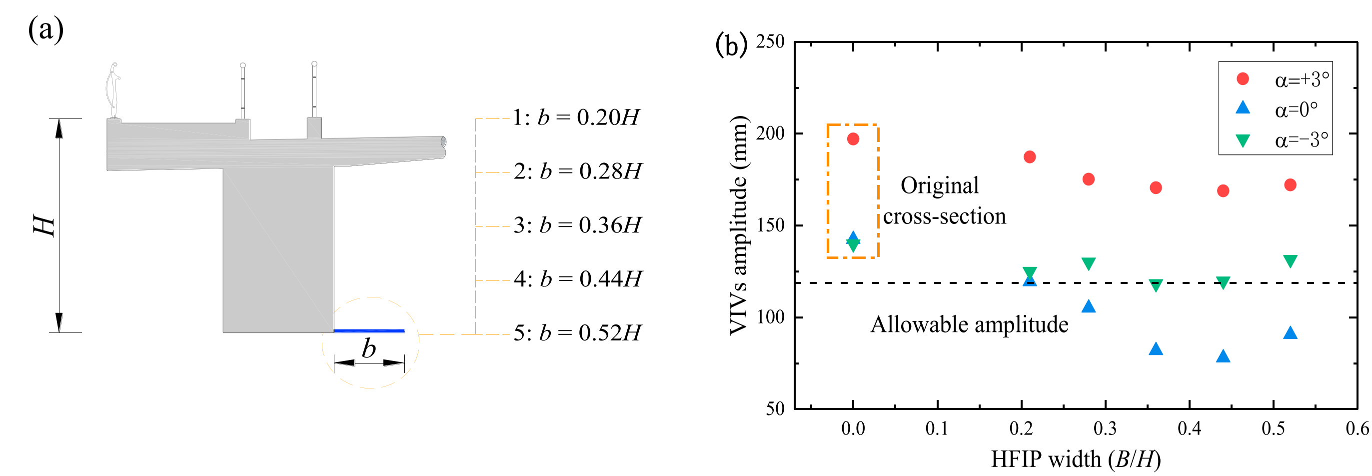


Fig. 15. Vertical VIV responses of deck with different HFIPs.

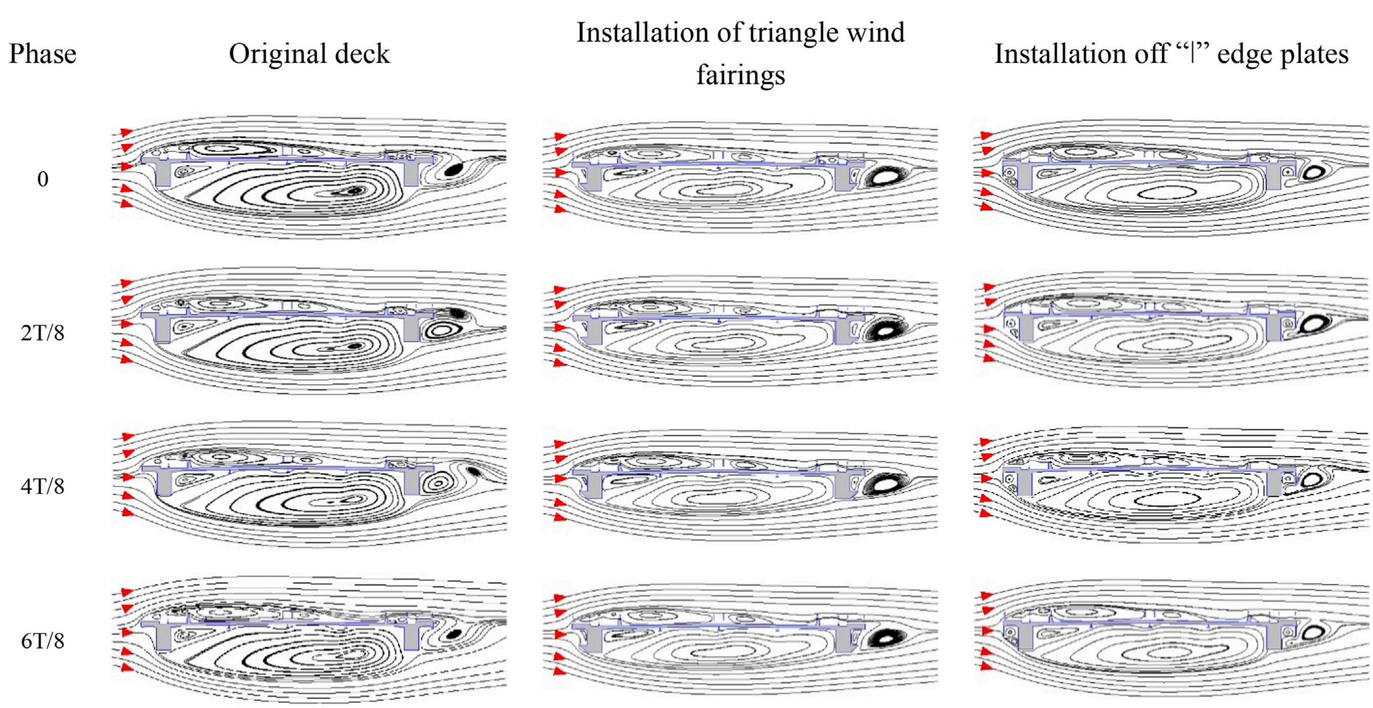


Fig. 16. Numerical simulation of improved bridge deck at 0° attack angle.

the alternating vortices have already disappeared. Moreover, compared with the results of the original deck, the form of the large-scale vortices under the deck also significantly change. The vortices are further away from the wake zone and are not affected by the shedding vortices. By installing the countermeasure of vertical "|" edge plate, a pair of small vortices is generated in the region between the windward edge plate and longitudinal box. The lower vortex is close to a triangle shape and locates on the windward side of the longitudinal box. It is interesting that the lower vortex is highly similar in the shape and location to the mini-triangular wind fairings. In addition, the analogous vortex structure can be found in the rear region of the deck at the 0° attack angle. These two countermeasures improve the section streamline type, which allows the incoming flow to pass over the section more smoothly, thereby decreasing both the scale and strength of vortices.

For the original deck, Fig. 17 shows that the dimensions of vortices at the upper and lower surfaces of the deck at the +3° attack angle are greater than those when the attack angle is 0°, the distortion of vortices is more severe, and the VIV amplitude is larger than that at 0° attack angle. When triangular wind fairings are installed on the edge boxes, the flow pattern in the numerical simulation at the +3° attack angle is similar to that at the 0° attack angle, the periodic vortex shedding in the wake zone practically disappears, and the large-scale vortices at the upper and lower decks are more stable, thereby suppressing the VIV. When the vertical "|" edge plate is installed when the attack angle is +3°, however, the periodic shedding vortices still exist but slightly diminish in intensity, and the distortion of vortices at the upper and lower sections relatively weaken synchronously.

Comparing with the flow patterns around the original cross-section at 0° and +3° attack angles, the large-scale vortices under the deck have roughly the same size, but the vortices at the attack angle of +3° distort more severely. For the upper region of the deck, the vortices at +3° attack angle have greater strength and larger scale than that at 0° attack angle. Evidently, the VIV amplitude of the deck at +3° attack angle is approximately 50% higher. It can be inferred that the instability of the vortices

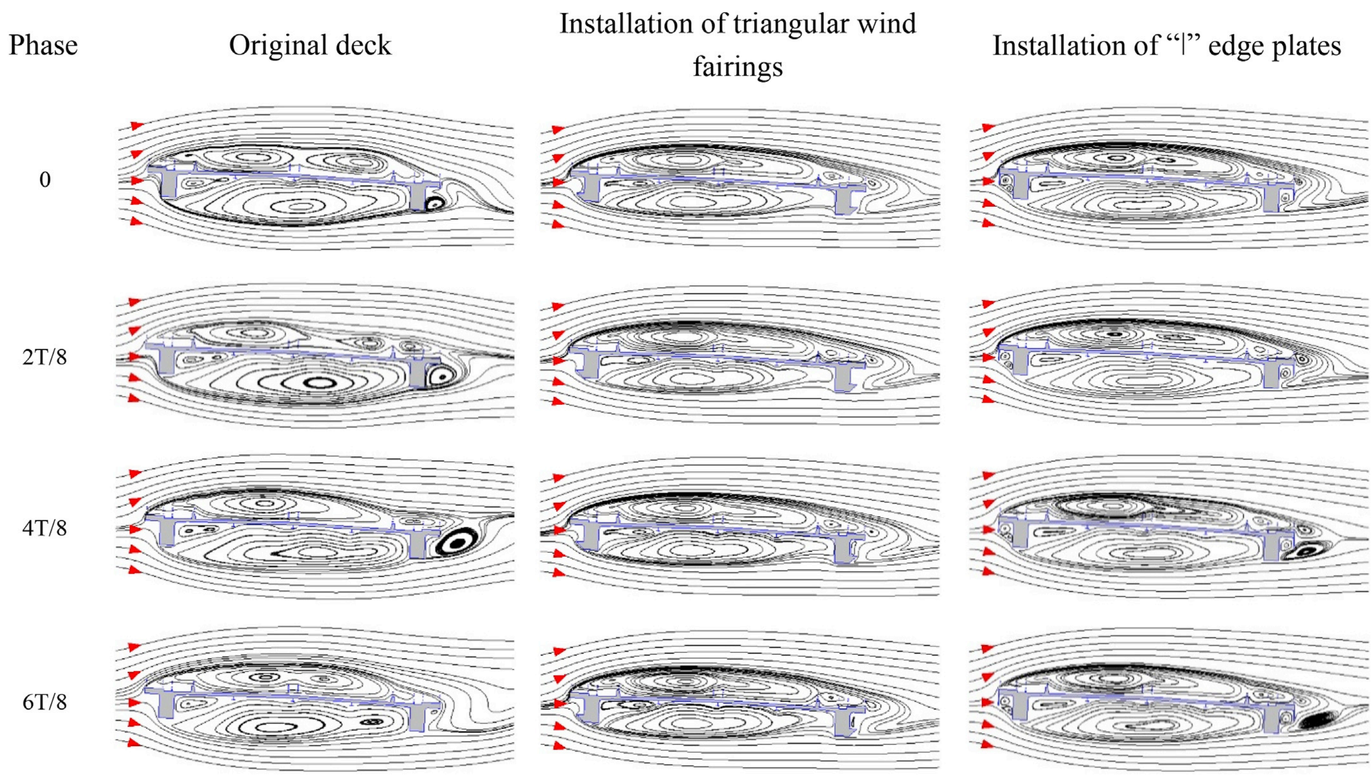


Fig. 17. Numerical simulation of improved bridge deck at $+3^\circ$ attack angle.

on the upper surface may be the probable reason for intensifying the VIV at $+3^\circ$ attack angle.

As illustrated in the streamline diagrams at two different attack angles, it can be seen that the installation of the triangular wind fairings would smooth the edges of the longitudinal box, which leads to a significant decrease in the flow separation at both the windward and leeward ends of the longitudinal box, consequently reducing the strengths of vortices under the deck and mitigating the periodic vortex-shedding in the rear region of the deck. Therefore, the aerodynamic interactions of the large-scale vortices under the deck and the shedding vortices are controlled, and suppressing the VIV ultimately. This is the probably reason for the VIV suppression of the installation of the triangular wind fairings. The "|" vertical plate, although successfully suppressing the VIV at 0° attack angle, does not work when the attack angle is $+3^\circ$. Compared to the flow pattern at 0° attack angle, periodic vortex shedding occurs in the rear region of the deck when the attack angle is $+3^\circ$. The separation at the windward longitudinal box is greater, and the vortices around the upper surface are unstable. As a result, the VIV is mitigated slightly.

5. Conclusions

In the present study, the VIV performance and aerodynamic mechanisms of the two-box edge girder cable-stayed bridge is investigated in detail using a series of section model wind tunnel tests and numerical simulations. Severe VIVs are observed in the wind tunnel tests at low wind speeds because of the bluff characteristics of the edge girder section. The flow pattern and vortex structure around the bridge deck under consideration at the 0° attack angle are numerically simulated to preliminarily understand the aerodynamic mechanism of the VIV. The separation of incoming flow at the windward side, the distortion of large-scale vortices under the deck, and the shedding vortices in the wake zone periodically would change the pressure on the deck surface and induce deck vibration. It could be inferred that three regions cause deck VIV: the windward region, the region under the beam, and the wake region.

Based on the mechanism analysis of the numerical simulation, several aerodynamic measures are designed, and their effects are initially investigated by a series of wind tunnel tests. The main findings are as follows. For the HFIP countermeasure, the most effective width to suppress the VIV is $0.44H$. With this width, the vertical VIV amplitude is reduced by 40% at the 0° attack angle and approximately 10% at the -3° and $+3^\circ$ attack angles. For the vertical "|" edge plate countermeasure, the VIV amplitude is significantly reduced at 0° and -3° attack angles; however, this countermeasure is not effective at the $+3^\circ$ attack angle. By using the mini-triangular wind fairing countermeasure ($s = 0.25H$; $\beta = 45^\circ$), the VIV responses significantly decrease at the three attack angles, thereby satisfying the requirement of the wind-resistant design standard.

Finally, the vertical "|" edge plate and triangular wind fairing are selected to analyze the aerodynamic mechanisms of countermeasures using the CFD method. After installing the triangular wind fairing with the attack angles at 0° and $+3^\circ$, the periodic vortex shedding in the wake zone practically disappears, and the large-scale vortices under the deck move further away from the wake zone. The severe distortion is also significantly weakened, thereby suppressing the VIV. In installing the "|" edge plate, it is found that the flow pattern around the section is similar to that when triangular wind fairings are installed at the 0° attack angle, thereby effectively suppressing the VIV. The periodically shedding vortices in the wake zone still exist at the $+3^\circ$ attack angle, and the distortions of large-scale vortices at the upper and lower sections slightly weaken, thereby reducing the VIV amplitude by only 10%. The tendencies of the VIV amplitudes in the section wind tunnel tests and the flow fields in the numerical simulations at different attack angles exhibit good agreement.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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